

SURROS - Role Wise Weekly Execution Plan (12 Weeks)

Week 1 – Research & Setup

Team Lead: Freeze risk definition & architecture draft.

Data Engineer: Collect raw datasets & inspect quality.

Backend Dev: Setup Django project & initial app structure.

Frontend Dev: Basic HTML layout skeleton.

Research Lead: Read 5 papers & prepare summary.

Week 2 – Data Cleaning & Maths Base

Team Lead: Implement normalization & risk formula.

Data Engineer: Clean dataset & monthly aggregation.

Backend Dev: Create DB models (Zone, RiskData, Prediction).

Frontend Dev: Build dashboard UI structure.

Research Lead: Document risk pillars & methodology.

Week 3 – OOP Structure

Team Lead: Create classes (DataLoader, RiskCalculator, SeasonalAnalyzer).

Data Engineer: Integrate processed data into DB.

Backend Dev: Build APIs (/zones, /risk/).

Frontend Dev: Connect dashboard with API.

Research Lead: Draw system architecture diagram.

Week 4 – Seasonal Analysis

Team Lead: Implement mean, variance & peak detection.

Data Engineer: Validate seasonal grouping accuracy.

Backend Dev: Seasonal API endpoint.

Frontend Dev: Monthly trend graph with Chart.js.

Research Lead: Document seasonal insight logic.

Week 5 – Prediction Module

Team Lead: Train Linear Regression & evaluate (RMSE, MAE).

Data Engineer: Prepare training/testing split datasets.

Backend Dev: Prediction API endpoint.

Frontend Dev: Display next month prediction result.

Research Lead: Write model explanation section.

Week 6 – Resource Recommendation

Team Lead: Implement rule-based resource allocation logic.

Data Engineer: Validate risk categorization thresholds.

Backend Dev: Resource recommendation API.

Frontend Dev: Resource suggestion panel UI.

Research Lead: Document allocation logic.

Week 7 – Admin Panel

Team Lead: Integrate retraining trigger.

Data Engineer: Create structured data upload format.

Backend Dev: Admin authentication & file upload handling.

Frontend Dev: Admin dashboard UI.

Research Lead: Document admin workflow.

Week 8 – Integration & Testing

Team Lead: System integration testing.

Data Engineer: Data validation & edge cases.

Backend Dev: API debugging & performance fix.

Frontend Dev: UI bug fixes.

Research Lead: Prepare test case documentation.

Week 9 – Advanced Insights

Team Lead: Correlation analysis & heatmap logic.

Data Engineer: Feature correlation dataset prep.

Backend Dev: Insights API endpoint.

Frontend Dev: Insights dashboard page.

Research Lead: Write insights interpretation.

Week 10 – Optimization

Team Lead: Code refactor & performance improvement.

Data Engineer: Optimize data queries.

Backend Dev: API speed optimization.

Frontend Dev: UI polish.

Research Lead: Update documentation.

Week 11 – Documentation & Report

Team Lead: Final technical validation.

Data Engineer: Dataset documentation.

Backend Dev: Deployment configuration.

Frontend Dev: Final UI cleanup.

Research Lead: Prepare final report & PPT.

Week 12 – Final Review & Demo

Team Lead: Full system review.

Data Engineer: Final dataset consistency check.

Backend Dev: Final deployment test.

Frontend Dev: Demo readiness check.

Research Lead: Final presentation rehearsal.